The Human Bindome

A proteome-scale atlas of designed binder candidates

Wenckstern, Díaz-Rovira et al. — bioRxiv, 2026

306,146 predicted candidates; 8,296 human proteins.
A catalogue of hypotheses awaiting experimental validation.

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No later journal version located in the citation check.

Can we make protein-level perturbation broadly accessible?

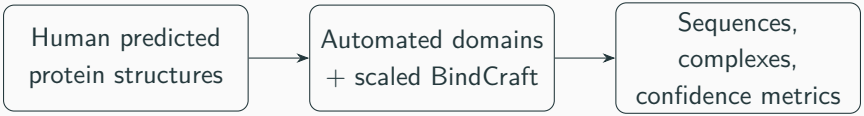
Genetic knockouts remove a protein; a binder can instead engage a particular surface.

- Block one interaction while preserving other functions.
- Mask a modification site or recruit an effector to a target.
- Detect, enrich, or stabilize a protein using an explicit sequence.

The bottleneck: discovery is usually target by target, and many antibody sequences and binding modes are unavailable.

Introduction and Discussion. Applications are motivations, not demonstrations in this atlas.

The contribution is a precomputed resource



New scale and access: automated target selection, efficient execution, proteome analyses, web/API/MCP access and ML splits.

Inherited design engine: BindCraft; the paper retains its default design and acceptance filters.

Fig. 1a; Online Methods: Proteome-scale binder generation.

What exactly is one entry?

Input	Canonical human target sequence, AlphaFold DB v6 structure and PAE matrix.
Design unit	A selected compact domain; no hotspot residues supplied.
Output	A 90-amino-acid binder sequence, predicted domain complex, full-target placement and metrics.
Identity	Target linked by UniProt accession or entry name.

An accepted entry means it passed computational filters. It does not establish binding affinity, specificity or activity.

Results: Proteome-scale binder design; Data Availability.

Step 1: turn full-length proteins into designable domains

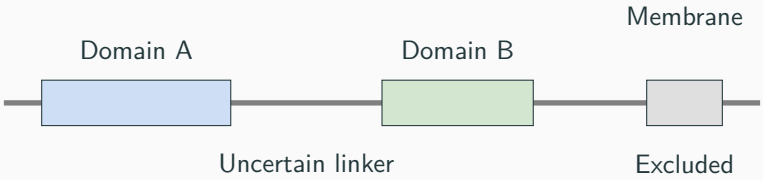
PAE estimates uncertainty in the relative positions of residues; graph clustering finds internally coherent regions.

- PAE clustering: resolution 1.0; cutoff 15 Å; extend boundaries by 2 residues, include gaps of at most 21 residues.
- Keep length 30–550 residues and mean pLDDT ≥ 0.7.
- Require compactness $R_g/N^{1/3} \leq 5.0$ and Cα contact density ≥ 2.0 within 8 Å.
- Exclude membrane-embedded regions and specified simple helix/loop-only domains.

These rules define the paper's *targetable proteome*; it is a workflow-dependent subset.

Online Methods: Domain segmentation and annotation; Target domain selection. Confidence values normalized to 0–1.

A running example: one protein, two plausible domains



Illustrative protein: design independently against A and B, each smaller than the complete target. A 90-aa binder that works geometrically on A must then be checked against B in the full model.

Domain cropping saves computation and explores multiple regions, but can expose surfaces that are inaccessible in the intact protein.

Toy schematic, not a reported protein. Rationale: Results and Online Methods.

Step 2: scale BindCraft without relaxing its filters

1. Optimize binder trajectories using the BindCraft framework.
2. Before ProteinMPNN sequence redesign, place the binder/domain complex into the full-length target and check clashes.
3. Redesign and assess candidates using the original BindCraft settings; generate up to 52 accepted designs per domain.

Engineering changes: independent SLURM array jobs, smaller target crops, and fixed 90-aa binders to avoid costly model recompilation.

The 90-aa length also accommodates downstream synthesis in 300-bp oligonucleotide pools.

Online Methods: Binder design. This deck summarizes the inherited engine rather than introducing a new design algorithm.

Step 3: reject clashes, but account for model uncertainty

After domain alignment to the full target, a clash with another well-positioned region leads to rejection.

Exception in the Methods
A clashing region can be tolerated when its maximum pLDDT is below 0.7 *or* its symmetrized PAE is above 15 Å.

Interpretation: uncertain coordinates should not automatically veto a candidate.

Remaining risk: the true cellular conformation may still occlude the site; oligomers, ligands and other proteins are not captured by this single-chain check.

Online Methods: Binder design; Extended Data Fig. 1. Caption says "low PAE" for a retained clash, inconsistent with the high-PAE Methods rule.

Confidence scores prioritize candidates; they do not validate them

Quantity	Acceptance threshold	Atlas median
Binder monomer pLDDT	> 0.8	0.92
Normalized interaction PAE	< 0.35	0.23

pLDDT measures predicted local structural confidence. Interaction PAE assesses uncertainty in binder–target placement.

The medians are away from their cutoffs; the full distributions and other retained BindCraft metrics appear in Extended Data Figs. 2–3.

Neither score is a measured dissociation constant or a selectivity assay.

Fig. 1b; Results: In silico confidence of the binder candidates.

Coverage depends strongly on the denominator

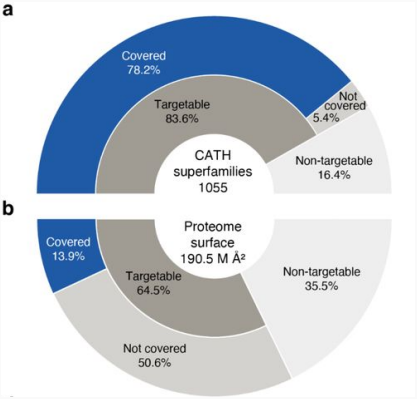
Paper-defined set	Human proteins
Reviewed human proteome	20,279
At least one targetable domain	10,664
At least one accepted candidate	8,296

40.9% of the full proteome; **77.8%** of the targetable proteome.

306,146 candidates: approximately 37 per covered protein and 30 per covered domain. An average is not a uniform allocation.

Fig. 1c and Results. Counts refer to manuscript v1; live portal reports 306,151 candidates for the same 8,296 targets.

Broad protein coverage still leaves sparse epitope coverage



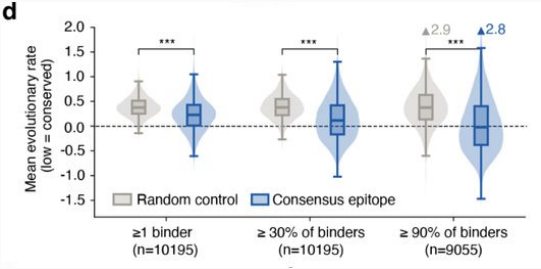
Structural families: 78.2% of 1,055 represented CATH superfamilies have at least one covered domain.

Surface: 64.5% of modelled proteome surface is targetable, but only 13.9% is contacted by candidates.

This is about 21.6% of targetable surface (13.9/64.5; derived). More candidates can revisit the same site.

Fig. 2a–b, cropped unchanged. Surface uses full-length-model SASA and maximum per-residue burial across binders.

Designed epitopes converge on conserved surfaces

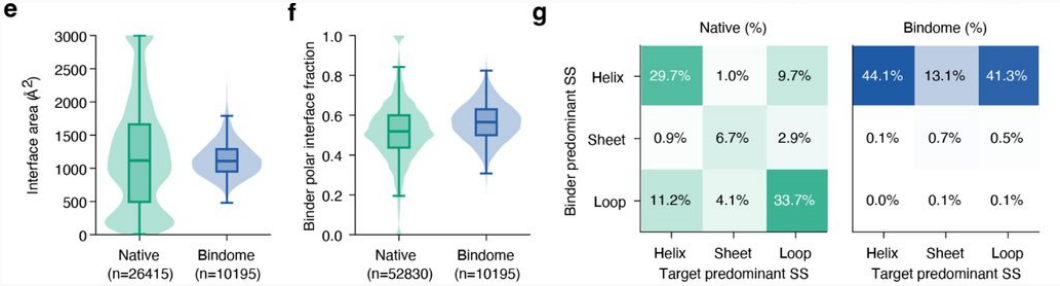


Consensus epitope: contact by any binder, $\geq 30\%$, or $\geq 90\%$ of a domain's binders.
Compare mean Rate4Site rate with a residue-count-matched surface control, paired by domain.
Lower rate means stronger conservation.
All three comparisons are significant; the trend strengthens with consensus.

Conservation supports functional relevance but also warns of cross-reactivity among paralogs. Significance alone does not quantify specificity.

Fig. 2d, cropped unchanged; Online Methods: Epitope evolutionary conservation analysis. Control pool may include epitope residues.

Interfaces resemble native contacts, with a strong helix bias



Interface size and polarity have narrower distributions than native PINDER interfaces. Binder interfaces are overwhelmingly helical: helix–helix 44.1%, helix–loop 41.3%, helix–sheet 13.1%.

Comparison uses one best-confidence design per domain; native complexes are filtered and deduplicated by interacting UniProt pair.

Fig. 2e–g, cropped unchanged; Online Methods: Bindome comparison to native protein-protein interactions.

Novelty and induced fit are prediction-based descriptors

- **Fold novelty:** 17.3% have maximum query TM-score < 0.5 against AlphaFold DB structures.
- **Interface novelty:** median best interface TM-score is about 0.3 against retrieved PPIRef matches.
- **Conformational adaptation:** interface Cα RMSD > 2 Å occurs for 4% of candidates and 8% of targets, comparing predicted unbound and bound structures.

These are not experimental measurements of new folds or induced fit. Search coverage and model biases influence the conclusions; target templating may suppress rearrangements.

Extended Data Figs. 5–6; Methods: Fold novelty, Interface novelty, Induced-fit signal.

Functional annotations turn entries into testable hypotheses

Target-level context: disease associations, Pharos development level, and essentiality in A375, K562 and HCT116.

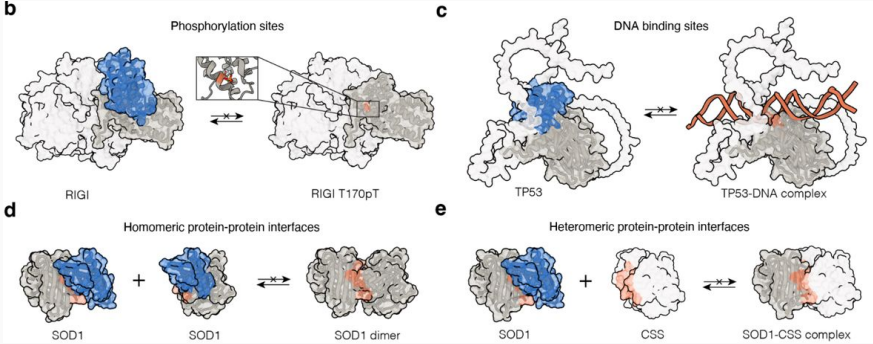
Site-level context: PTMs, native protein interfaces, and ligand-binding residues.

- 148 targets have candidates overlapping annotated DNA-binding regions.
- Roughly half of annotated homo-/heteromeric interfaces overlap a candidate epitope for a partner.
- Around one quarter of targets with ligand-binding annotations have an overlapping candidate.

The overlap rule can be as permissive as one shared residue; “disrupting” is an annotation-based nomination.

Fig. 3a; Methods: Proteome-scale nomination of perturbagen candidates. Essentiality: Chronos gene effect < -1.0.

Four proposed perturbations, not four demonstrated effects



RIG-I: mask a phosphorylation region. TP53: compete with DNA.
SOD1: interfere with dimerization or the copper chaperone CCS.

Fig. 3b–e, cropped unchanged. Blue: designed candidate; orange: functional site/partner contact. Arrows depict hypotheses, not measured equilibria.

Allosteric overlap: an interesting claim needing clarification

Reported result: 67 of 256 human ASD targets (26%) have a candidate overlapping an allosteric site; examples include AKT1, TEAD2 and DUSP10.

The dedicated overlap method maps modulator-contact residues through SIFTS and tests shared residues with binder interfaces.

Internal inconsistency: a later Methods paragraph says ASD sites were *not* resolved to residues and reports only whole-target annotation.

Treat the 26% site-overlap claim as author-reported, with a reproducibility question. Even verified spatial overlap would not establish allosteric modulation.

Fig. 2h–i; Methods: Coverage of allosteric sites versus later Allosteric-site coverage paragraph.

A practical path from the atlas to an experiment

1. Query a target such as TP53 (P04637) on bindome.epfl.ch.
2. Rank candidates by confidence, then inspect different epitopes in the full target context.
3. Download sequences and models; select diverse candidates rather than only many variants of one backbone.
4. Test expression, binding and specificity; assay DNA competition or the intended cellular phenotype.

Suggested controls: unrelated binder, closest paralog/off-target panel, dose response, and independent candidates engaging the same site.

Fig. 4 and server workflow. Experimental selection and controls are journal-club proposals, not experiments reported here.

ML splits aim to prevent interface memorization

Unit of splitting: paired target-interface and binder-interface clusters, built from interface-filtered Foldseek alignments.

All members of a paired cluster enter one partition. Depth-2 graph deleaking removes training examples close to held-out complexes.

Training	Validation	Test
205,553	21,205	21,955

Usable total: 248,713 complexes (81.2% of input). After deleaking, 23 of 88,720 multi-member backbone groups still cross training/holdout partitions.

Fig. 4d; Methods: Leakage-controlled data splits. Labels remain model-generated, not experimental binding outcomes.

What is established, and what should journal club challenge?

Established by the computational analyses
A large, open, sequence-defined catalogue with broad target coverage, predicted epitopes and useful access infrastructure.

Unresolved biological generalization
Atlas-wide binding success, affinity, solubility, intracellular stability, paralog selectivity and functional perturbation remain unmeasured here.

1. How should a representative validation panel sample targets and epitopes?
2. Should future progress prioritize more proteins or more distinct surfaces?
3. How much apparent confidence survives independent models and assays?

Discussion; journal-club interpretation. Prior BindCraft success rates cannot be transferred unchanged to this atlas.

Sources and version discipline

Paper read: Wenckstern, Díaz-Rovira et al., *The Human Bindome: A Proteome-scale Atlas of Designed Binder Candidates*. bioRxiv v1, 30 July 2026. DOI: 10.64898/2026.07.30.741542.

Methods and supplement: complete online text, Online Methods and Extended Data Figs. 1–6 captions. Main Figures 2 and 3 reproduced as labelled crops under CC BY 4.0.

Resource checks: `https://bindome.epfl.ch`; `https://github.com/wejulius/Bindome`; dataset: `https://hf.co/datasets/wjulius/HumanBindome`.

Citation checked 1 November 2026. Portal count is 306,151 versus 306,146 in v1; use manuscript counts for the reported analyses. Methods inconsistencies are flagged on the relevant slides.